

Rydberg Physics in Quantum Microwave-Optical Converters: A Qualitative Analysis

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Abstract

Following the development of quantum computers since the twentieth century, the transportation of quantum information encoded in microwave photons has emerged as a significant challenge and viable solutions are still under active development. While long-distance optical quantum communication has achieved considerable success, this motivates efforts to bridge the microwave and optical regimes through the development of microwave-optical converters. This thesis provides a qualitative review of the state-of-the-art development of microwave-optical converters based on Rydberg platforms. The physical properties of Rydberg atoms and Rydberg excitons that enable strong coupling to microwave fields are discussed, together with experimental implementations of microwave-optical conversion using ultracold and room-temperature Rydberg atoms. For completeness, several non-Rydberg platforms are also briefly reviewed and compared.

Supervisor: Peter Swift

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2 Introduction

The earliest investigations of Rydberg physics date back to 1934, when Amaldi and Segrè examined pressure-induced shifts and spectral broadening of highly excited states in sodium and potassium atoms arising from collisions with rare gases [1]. Since then, the unique properties of Rydberg atoms and Rydberg excitons such as their large electric dipole moments and strong interactions, have been extensively explored and harnessed in quantum technologies. These include quantum computing [2–6], quantum sensing [4, 5, 7], and microwave–optical conversion [8–11].

With the rapid development of quantum computing throughout the twentieth and twenty-first centuries, significant effort has been devoted to microwave-optical converters (MOCs). Such devices are essential for coherently transferring quantum information encoded in microwave photons into optical photons suitable for long-distance transmission.

In this thesis, we qualitatively review the development of MOCs based on Rydberg platforms and analyse the limitations and performance characteristics of each approach.

This thesis is structured as follows. In Section 3, we discuss the motivation for developing microwave-optical converters and review several non-Rydberg approaches. In Section 4, we introduce the Rydberg platforms that can be used for MOC implementations. These platforms are examined in detail in Section 5 and Section 6. Finally, in Section 7, we discuss possible directions for improving MOCs based on Rydberg platforms and the potential of Rydberg excitons for future MOC development.

3 Background

3.1 Development of Quantum Computing

In 1936, Alan Turing introduced a theoretical model of computation based on algorithms operating on a binary logic system [12], laying the foundation of classical computing. In modern computers, this binary logic is physically realised using Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFETs) [13], in which the flow of current is controlled by an applied gate voltage to represent logical states. Large numbers of these transistors are integrated into multi-core central and graphics processing units, enabling parallel processing of computational tasks [14].

Despite these advances, the computational time required by classical computers to solve certain problems such as integer prime factorisation and the discrete logarithm problem scales exponentially with problem size [15, 16]. This fundamental limitation motivates the search for computational paradigms capable of outperforming classical computers for specific classes of problems.

With the development of quantum mechanics in the twentieth century, physicists began to explore whether uniquely quantum phenomena could be exploited to overcome the limitations of classical computation discussed above. In particular, a quantum computer can, in principle, solve the prime factorisation problem in polynomial time using Shor’s algorithm, in stark contrast to the exponential scaling encountered in the best-known classical algorithms [15].

Several key milestones in the twentieth century laid the groundwork for the physical realisation of quantum computation. In 1953, Wolfgang Pauli demonstrated that charged particles could be trapped using time-varying electric fields, circumventing the constraints imposed by Earnshaw’s theorem [17, 18]. In 1985, Devoret, Martinis and Clarke observed macroscopic quantum tunnelling of Cooper pairs in Josephson junctions [19], establishing that quantum coherence could be engineered and measured in macroscopic solid-state systems. Around the same period, Ekimov, Brus and Bawendi independently developed semiconductor nanostructures - quantum dots that exhibit quantum confinement effects [20]. These discoveries enabled the development of the fundamental units for quantum computers - qubits [6, 21–23] and quantum information processing based on these platforms has since been successfully demonstrated experimentally [3].

3.2 Motivation of Developing Microwave-Optical Converters

3.2.1 Technical Challenges of Quantum Communication

The remarkable progress of classical computing in the modern world is largely due to the ability of computers to communicate efficiently over local area networks (LANs) and the Internet. For quantum computers to achieve a comparable level of scalability and practical utility, it is neces-

sary to develop quantum networks that enable distributed quantum processors to communicate with one another. This can be achieved through the exchange of quantum information between quantum computers.

As discussed in Section 3.1, three types of qubits have been introduced, among which superconducting circuits and semiconductor quantum dots operate in the microwave frequency regime [6, 8]. A key challenge arises when transporting quantum information encoded in microwave photons over long distances, as it is highly susceptible to dissipation and decoherence due to thermal noise [8, 24].

Quantitatively, microwave photons at frequencies around 10 GHz experience losses in low-loss microwave cables that are approximately 10^4 times greater than those of optical photons propagating in optical fibres at telecommunication wavelengths [24]. Consequently, direct microwave transmission is impractical for long-distance quantum communication. This motivates the use of a technically mature optical communication infrastructure as an alternative medium for transmitting quantum information.

3.2.2 Achievements in Optical Quantum Communication

The development of long-distance optical quantum communication, in which quantum information is encoded in optical photons, has achieved significant progress in the twenty-first century.

One notable example of long-distance quantum communication is quantum key distribution (QKD). QKD is a secure communication protocol that enables two distant parties to share a common secret key for encrypting and decrypting messages [25]. Terrestrial free-space QKD was demonstrated by Liao *et al.* (2017), who successfully generated and transmitted secret keys at kilohertz rates from a satellite to a ground station over a distance of approximately 1200 km [25]. Furthermore, fibre-based QKD was achieved by Boaron *et al.* (2018), who demonstrated secret key distribution over optical fibre at distances of up to 421 km [26].

Beyond quantum key distribution, long-distance optical quantum communication has also been realised through the distribution of entanglement between matter-based quantum systems mediated by photons. Yu *et al.* (2020) demonstrated entanglement between two distant atomic ensembles separated by 22 km using two-photon interference, and up to 50 km using single-photon interference [27]. Such photon-mediated atom-atom entanglement represents a key building block for scalable quantum networks, enabling the coherent transfer of quantum information between stationary quantum memories and travelling photonic qubits.

Motivated by the maturity of optical quantum communication technologies, significant effort has been devoted to developing interfaces that coherently convert quantum information between microwave and optical frequency regimes. Such microwave-optical converters (MOCs) would enable quantum information processed in microwave-based quantum computers to be preserved and transmitted over long distances using low-loss optical channels. In the section below, we discuss briefly the non-Rydberg approach in the fabrication of MOCs.

3.3 Microwave-Optical Conversion: Non-Rydberg Platforms

Several non-Rydberg platforms were proposed as the microwave-optical converters such as opto-electro-mechanical transducer [28], opto-mechanical [24, 29], piezoelectric [24, 29] and magnons [24, 30].

To be considered a viable microwave-optical converter, a device must exhibit both high conversion efficiency, scalability to large-scale fabrication and bandwidth. In practice, however, these requirements are often in tension, as designs that achieve high efficiency tend to be difficult to manufacture at scale and have smaller bandwidth, while scalable platforms typically suffer from reduced coupling strength. In the sections below, we will explore a few examples.

3.3.1 Opto-Electro-Mechanical Microwave-Optical Converters

Nano-scale opto-electro-mechanical converters have been widely studied [28, 31–34] and are considered a promising platform for quantum communication.

A basic opto-electro-mechanical MOC consists of a pair of capacitor pads inside a Fabry-Pérot cavity, with one embedded in a thin membrane and the other integrated into an inductor-capacitor (LC) circuit [28, 33, 35]. When the inductor picks up microwave photons from the coupling port or external microwave radiation, the magnetic energy of the photons is converted into electrical energy, charging the capacitor [36]. As the electromagnetic field oscillates, the capacitor exerts a time-varying force on the membrane pad, inducing mechanical vibration [35]. This mechanical vibration effectively changes the length of the cavity and hence modifies both the cavity resonance and the optical frequency [34].

The general setup described above is adapted in different experiments with variations in materials and biasing methods. For example, in [35], the capacitor pair is composed of aluminium-coated silicon nitride and a four-segment gold electrode, while in [28], the membrane is silicon nitride with embedded niobium, coupled to a microfabricated niobium circuit. Biasing is achieved via a voltage in [35] and through strong microwave or optical pumps in [28].

To date, [28] reports the highest conversion efficiency among microwave-optical converters (MOCs), reaching up to 47% [28, 29], with minimal added noise. However, despite the high efficiency enabled by the Fabry-Pérot cavity, which requires extremely precise alignment, such devices are generally difficult to mass-produce [29]. Furthermore, the reported bandwidth of 1.2×10^{-2} MHz [28, 34] is relatively narrow compared with magnon-based MOCs, as discussed below.

3.3.2 Ferromagnetic Magnons

As the temperature increases above 0 K, thermal fluctuations cause misalignment of magnetic moments in a ferromagnet, thereby reducing the net magnetisation. To achieve the minimal energy configuration, the magnetic moments precess collectively about their equilibrium positions,

forming spin waves. Upon quantisation, these spin waves give rise to quasiparticles known as magnons [37].

As the magnetic moments precess coherently and in phase, the net magnetisation is maximised. Such magnetostatic mode is referred to as the Kittel mode [30]. Owing to its large magnetic dipole moment, this mode couples strongly to microwave photons confined in a cavity [30, 38]. Aside from MOC, this mode has also been employed in demonstrations of quantum entanglement [39].

In the experiment reported in [30], a ferromagnetic sphere made of yttrium iron garnet (YIG) is placed inside a microwave cavity, where magnons are driven by microwave photons. The cavity is positioned within a static magnetic field provided by a permanent magnet, which magnetises the YIG sphere and induces circular birefringence via the Faraday effect [40]. As a result, the precessing magnetisation associated with the excited magnons modulates the polarisation of the laser field incident on the YIG at the magnon frequency ω_m . This time-dependent modulation generates optical sidebands at angular frequencies $\Omega_L \pm \omega_m$, where Ω_L denotes the angular frequency of the incident laser.

Although magnon-based MOCs exhibit relatively large bandwidths on the order of 1 MHz, their conversion efficiency remains very low, approximately $10^{-8}\%$ [30, 34], primarily due to the intrinsically weak magnon-light coupling rate.

As discussed above, non-Rydberg MOC platforms continue to face technical challenges in simultaneously achieving high conversion efficiency and large-scale scalability. We now shift our focus to Rydberg-based platforms and qualitatively review the development of Rydberg atoms and Rydberg excitons for microwave-optical conversion.

4 Rydberg Platforms

4.1 Introduction to Rydberg Atoms

Extended discussions of the properties of Rydberg atoms can be found in [1, 3, 5, 41]. In this thesis, we summarise several relevant properties and discuss how they can be utilised in MOCs based on Rydberg platforms.

Rydberg atoms are atoms in which the valence electron is excited to a very high principal quantum number n [41, 42]. Previous experiments have successfully excited caesium and rubidium atoms to $n = 45$ and $n = 50$, respectively [43, 44]. Because of this, Rydberg atoms are often modelled using the hydrogenic approximation for simplicity, even though they are multi-electron systems in which screening effects from inner electrons are present [5, 41].

Such high values of n lead to exceptionally large atomic radii, which scale as $\langle r \rangle \propto n^2$ for low orbital quantum numbers L [1, 41]. As a numerical example, in the ideal case where quantum defects are negligible, the atomic radius of the $45S_{1/2}$ state of caesium is approximately $0.13 \mu\text{m}$, compared with about 79 pm in its ground state.

Owing to their large spatial extent, Rydberg atoms are extremely sensitive to external perturbations, particularly electric fields [5, 45]. This sensitivity is quantified by the atomic polarisability, which scales as $\alpha \propto n^7$ [5, 41], as predicted by second-order perturbation theory. This strong scaling underpins the use of Rydberg atoms in quantum sensing technologies. To understand their applicability in microwave detection, it is therefore important to consider the frequency range to which Rydberg atoms are sensitive.

The energy levels of Rydberg atoms scale as $E_n \propto n^{-2}$ [5, 41]. As n increases, the energy separation between neighbouring states decreases rapidly, giving rise to transition frequencies that lie in the radio-frequency (RF) to terahertz regimes, depending on the specific Rydberg states involved [42, 46]. Consequently, Rydberg atoms are well suited for electromagnetic field detection. This property enables Rydberg atoms to detect the electric field of microwave photons and convert it into an optical field.

Different atomic species have been employed in various Rydberg atom experiments. For example, helium Rydberg atoms have been used in studies of Rydberg-Stark deceleration [47], while strontium atoms have been utilised in the production of circular Rydberg states [48]. For microwave-optical conversion, however, alkali metals such as rubidium [10, 11] and caesium [49] are commonly used. This is because their atomic structure is relatively simple and their excitation frequencies are well characterised [18]. In addition, the single valence electron in alkali atoms is weakly bound and becomes spatially extended when excited to high principal quantum numbers, making these atoms particularly sensitive to the electric fields of microwave photons.

4.2 Introduction to Rydberg Excitons

In solids, the overlap of atomic valence electron wavefunctions leads to the formation of valence bands (VB) and conduction bands (CB) [50]. In semiconductors, the VB and CB are separated by a forbidden energy gap containing no electronic states. As a result, interband transitions can occur only when valence electrons acquire energy greater than or equal to the band gap.

When a valence electron is excited into the conduction band, it leaves behind a positively charged hole in the valence band. The electron and hole interact via the Coulomb force and become bound to each other, forming a quasiparticle known as an exciton. Excitons, as the counterpart of their atoms with similar properties, possess discrete energy levels within the band gap [42] (as shown in Fig. 1) and can be excited to high principal quantum numbers, forming so-called Rydberg excitons.

Here, we restrict the energy dispersion to the region around the Γ point, where the effective masses of the electron and exciton in the crystal are positive under the effective mass approximation [37, 42, 51]. A direct consequence of the effective mass approximation is a parabolic band structure (as shown in Fig. 1), which arises from the Taylor expansion of the energy dispersion around the band minimum.

Since excitons can be readily destroyed by thermal ionisation at elevated temperatures, not all materials are suitable hosts for Rydberg excitons. In materials with small exciton binding energies, excitons cannot survive even at relatively low temperatures. The maximum observable excitonic state in a given material is given by [41]

$$n_{\max} = \sqrt{\frac{E_B}{k_B T}}, \quad (1)$$

where E_B is the exciton binding energy. For a conventional semiconductor such as GaAs, which has a low exciton binding energy of 4.2 meV [52], the maximum observable state at $T = 4$ K (the temperature of liquid helium) is only $n = 3$. Excitonic states with $n > 3$ are therefore destroyed by thermal effects. To maximise the highest observable excitonic state, a material with a large exciton binding energy is required.

In recent years, cuprous oxide (Cu_2O) has been proposed as a promising alternative to conventional semiconductors owing to its unusually large exciton binding energy of approximately 90 meV [41, 53]. Under millikelvin conditions, excitonic states with principal quantum numbers as high as $n = 30$ have been observed in Cu_2O [54], which remains the highest excitonic state reported in any semiconductor to date.

The large exciton binding energy in Cu_2O can be understood from the modified Bohr model for excitons [50, 55]:

$$E_B = -\frac{\mu e^4}{32\pi^2 \hbar^2 \epsilon_0^2 \epsilon_r^2} \frac{1}{n^2}, \quad (2)$$

where μ is the reduced mass of the electron-hole pair and ϵ_r is the relative dielectric constant

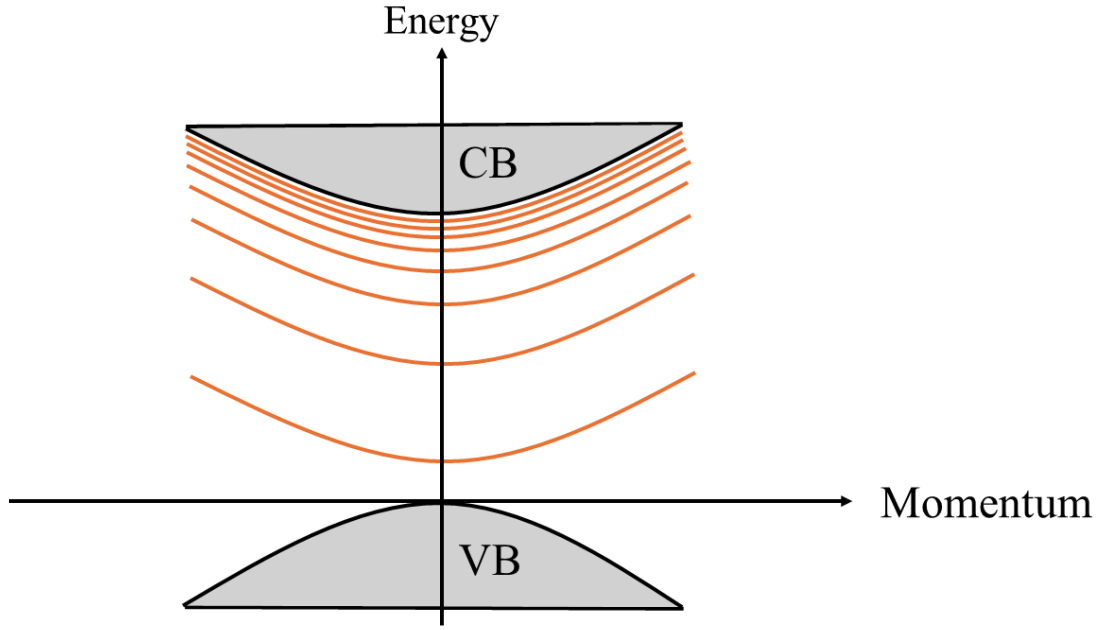


Fig. 1: Schematic of the energy levels of excitons. When an electron is excited from the valence band (VB) to the conduction band (CB), it leaves behind a positively charged hole in the VB. The electron and hole become bound via Coulomb attraction, forming a quasiparticle known as an exciton. The energy of an exciton can be illustrated as discrete levels within the band gap between the VB and CB.

of the host material. Compared to GaAs, Cu_2O exhibits a significantly larger reduced mass [55–57] as well as a smaller dielectric constant [50, 57, 58], implying a larger E_B of Cu_2O .

5 Microwave-Optical Converter via Rydberg Atoms

5.1 Six-Wave Mixing of Ultracold Rydberg Atoms

Han *et al.* (2018) realised a six-wave-mixing microwave-optical converter by forming a closed interaction loop consisting of four auxiliary optical fields and two microwave fields [10]. Six-wave mixing is a fifth-order nonlinear optical process ($\chi^{(5)}$), in which five input fields interact coherently within a medium to generate a new optical field (shown in Fig. 2), whose frequency satisfies both energy conservation and phase-matching conditions [16, 59].

In this scheme, the probe and coupling lasers, denoted by fields P and C, respectively, are directed into a cold ^{87}Rb atomic cloud at a temperature of $70\text{ }\mu\text{K}$, forming a Rydberg electromagnetically induced transparency (EIT) configuration [60, 61]. When the auxiliary fields R, M and A are applied, a closed-loop level structure is formed, as shown in Fig. 3(a). The microwave fields A and M couple the Rydberg state $|3\rangle$ to neighbouring Rydberg states $|4\rangle$ and $|5\rangle$, respectively. The optical field R then drives the transition from $|5\rangle$ to $|6\rangle$ [18].

Subsequently, the generated optical field L emerges through coherent wave mixing as the atomic population cycles through the closed loop. Importantly, this emitted field is not the result of spontaneous decay from $|6\rangle$ to the ground state, but rather arises from a coherent nonlinear optical process governed by the frequency- and phase-matched six-wave mixing interaction, described by

$$\omega_L = \omega_P + \omega_C - \omega_A + \omega_M - \omega_R \quad (3)$$

$$\mathbf{k}_L = \mathbf{k}_P + \mathbf{k}_C - \mathbf{k}_A + \mathbf{k}_M - \mathbf{k}_R \quad (4)$$

In the absence of the microwave field M, the loop is broken and the system reduces to microwave-dressed Rydberg EIT, in which no frequency-converted optical field is generated.

For applications in quantum information processing, it is essential that microwave-to-optical conversion preserves quantum coherence [16], so that phase and amplitude information are not degraded during the conversion process. To verify coherence, the authors employed optical heterodyne detection, in which the converted optical field L is superimposed with a local oscillator that is phase-locked to the probe field P. The presence of a stable and well-defined interference pattern indicates phase coherence between the fields [40].

Experimentally, a uniform $|\text{sinc}|$ spectral profile was observed, which is consistent with coherent frequency conversion and confirms that the phase information is preserved during the microwave-to-optical conversion process.

A further verification of coherence was performed by phase-modulating the microwave field M with a triangular waveform. The successful recovery of this modulation in the heterodyne signal of the converted optical field demonstrates faithful phase transfer from the microwave domain to the optical domain, confirming coherent conversion.

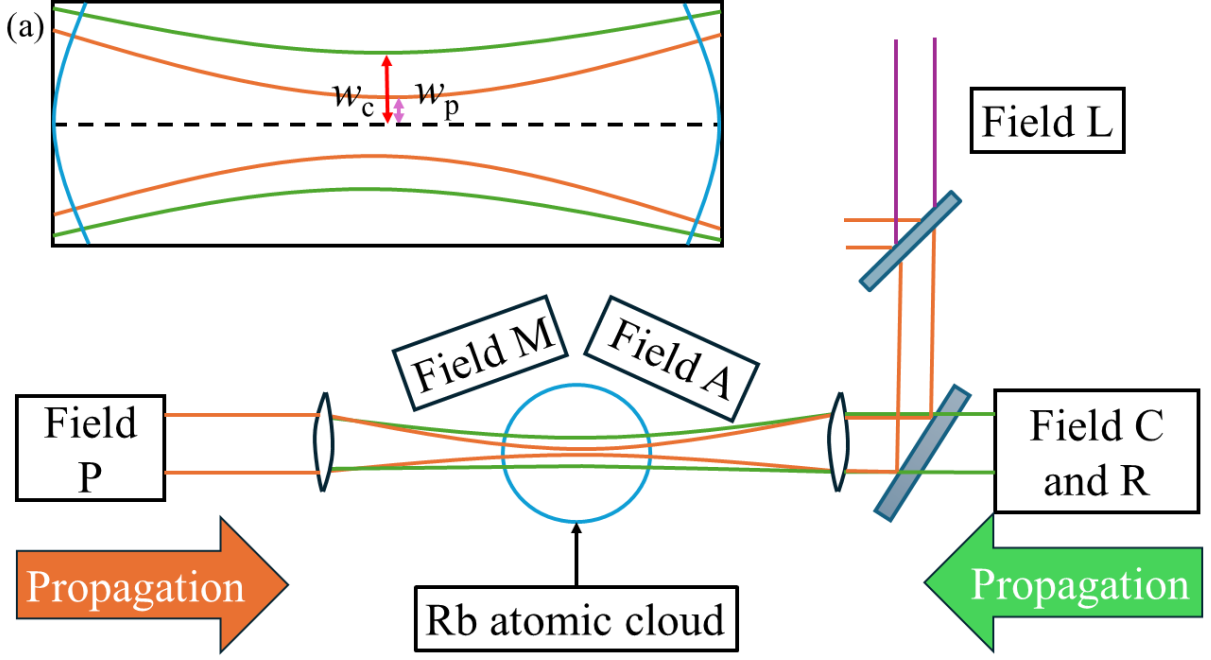


Fig. 2: A simplified diagram of the experimental setup for microwave-optical conversion using rubidium Rydberg atoms. The orange lines and arrow indicate the light path and propagation direction of field P, while the green lines and arrows indicate those of fields C and R. The inset (a) shows a zoomed-in view of the Rb atomic cloud, where w_c (red double-headed arrow) and w_p (pink double-headed arrow) denote the beam waists of fields C and P, respectively. The beam waists occur at the same location but are shifted slightly for clarity in the figure. The six-wave mixing process generates the field L, whose propagation path is indicated in purple.

This microwave–optical conversion (MOC) system initially achieved a conversion efficiency of approximately 0.3% with an exceptionally large bandwidth of 4 MHz. The same group later improved the efficiency to about 5% [11] by reconfiguring the geometry such that the microwave field M was aligned parallel to the propagation direction of field P.

The enhancement was attributed to improved spatial mode overlap with the atomic ensemble. The atomic cloud has a Gaussian spatial distribution centred in the interaction region, characterised by a waist w_z . The conversion efficiency depends on the effective number of atoms participating in the six-wave mixing process, which in turn is determined by the spatial overlap between the microwave field and the optical probe field. The probe beam has a waist [40] w_p , with $w_p \ll w_z$.

In the perpendicular configuration, the interaction volume is restricted by the transverse overlap between the microwave field and the tightly focused probe beam, effectively limiting the participating atomic cross-section to a region on the order of $2w_p$. In contrast, when the microwave field is aligned parallel to the probe beam, the interaction length extends across the full longitudinal extent of the atomic cloud, on the order of $2w_z$. This substantially increases the effective interaction volume and the number of atoms contributing to the nonlinear process, thereby enhancing the conversion efficiency.

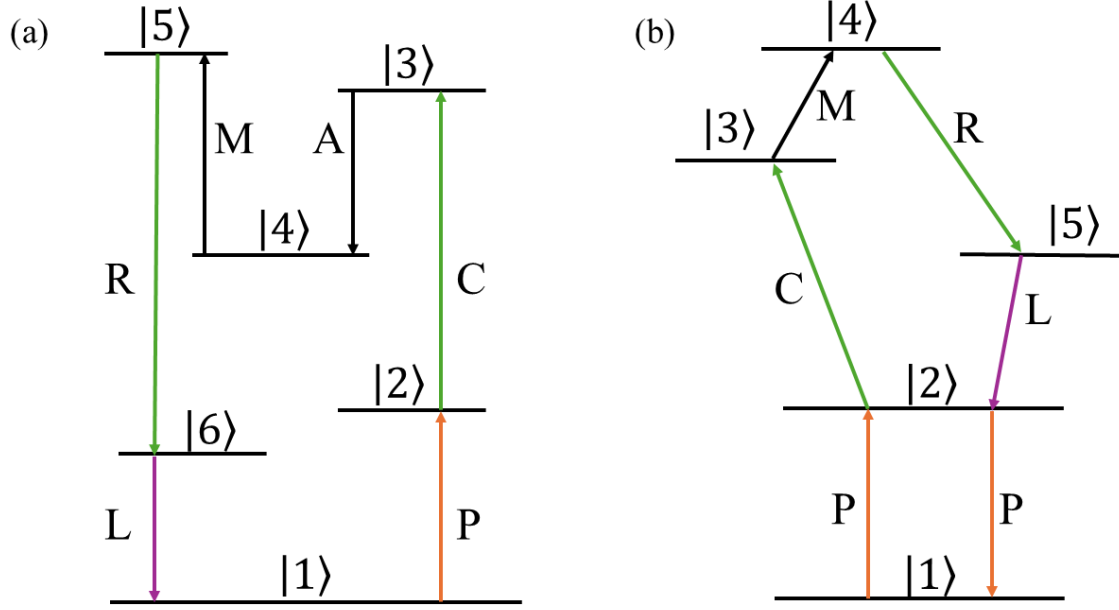


Fig. 3: A comparison of the energy levels involved in microwave-optical converters (MOCs) based on a closed-loop, six-wave mixing scheme using rubidium Rydberg atoms. The colour coding of the arrows is consistent with Fig. 2. (a) Energy levels used in the MOC implementation with ultracold rubidium atoms. (b) Energy levels used in the MOC implementation with room-temperature rubidium atoms. Here, field A is replaced by field P to facilitate the transition from $|2\rangle$ to $|1\rangle$.

In the following subsection, we will discuss how this setup can be simplified, specifically, by removing the cooling process and one of the microwave field used in [10, 11].

5.2 Six-Wave Mixing of Room Temperature Rydberg Atoms

Borówka *et al.* (2023) demonstrated microwave-optical conversion using Rydberg atoms at room temperature [62]. Their setup is similar to the scheme discussed in Section 5, except that the laser cooling stage was removed and only a single microwave field was applied (as shown in Fig. 3(b)). Using this configuration, they successfully converted a 19 GHz microwave signal into a 776 nm near-infrared optical field.

The performance of the converter is notable, achieving an efficiency of up to 2.8% and a bandwidth of 16 MHz. One reason for the significantly larger bandwidth compared with ultracold Rydberg atom systems is Doppler broadening, which causes the atomic energy levels to become spectrally broadened at room temperature. However, the higher temperature also reduces the lifetime of the Rydberg states. Consequently, the dephasing (decoherence) rate of room-temperature Rydberg atoms is approximately 10^3 times greater than that of ultracold Rydberg atoms, corresponding to coherence times on the order of nanoseconds at room temperature compared with microseconds in ultracold systems [5, 63, 64].

Furthermore, such a setup requires multiple laser systems arranged on an optical table

to excite the Rydberg states and realise Rydberg-dressed EIT. In contrast, superconducting qubits operate inside dilution refrigerators at millikelvin temperatures [8, 42]. As a result, the experimental infrastructure required for both ultracold and room-temperature Rydberg atom platforms is fundamentally incompatible with superconducting qubit architectures.

In addition, improving the microwave sensitivity of Rydberg atom-based MOCs typically requires excitation to higher principal quantum numbers n , which often necessitates additional laser stages and further increases experimental complexity.

These limitations motivate the search for a solid-state platform that retains the key properties of highly excited Rydberg atoms while operating at comparatively lower excitation levels. Such a platform would allow direct integration within a dilution refrigerator, enabling microwave-optical conversion in close proximity to superconducting qubits.

6 Coupling between Rydberg Excitons and Microwave Field

Rydberg excitons have emerged as a promising alternative to Rydberg atoms. Unlike atomic systems, excitons inherently exist within semiconductors [41], making them naturally compatible with solid-state platforms. This property renders them attractive candidates for integration with superconducting quantum circuits. Furthermore, even at comparatively low principal quantum numbers, Rydberg excitons exhibit characteristic features analogous to highly excited Rydberg atoms as illustrated in Table 1 [65]. These properties enable strong coupling to microwave fields while maintaining compatibility with cryogenic environments.

Quantity	Yellow, Cu ₂ O ($n = 25$)	Cs ($n = 80$)	Units
Binding energy (R_y/n^2)	0.0015	0.0017	eV
Orbital/atomic radius (r)	1030	980	nm

Table 1: Comparison of the properties of yellow Cu₂O excitons ($n = 25$) and Cs Rydberg atoms ($n = 80$), adapted from [65]. The yellow Rydberg excitons in Cu₂O at $n = 25$ exhibit properties comparable to those of caesium Rydberg atoms at $n = 80$, which can significantly simplify the experimental setup, particularly the laser and cooling systems.

To date, no fully realised microwave-optical converters based on Rydberg excitons have been experimentally demonstrated. The development of Rydberg exciton-based MOCs therefore remains at an early stage. Current literature primarily focuses on the coupling between Rydberg excitons and microwave fields [8], and the corresponding experimental investigation is discussed below.

6.1 One-Photon Absorption Spectroscopy

One-photon absorption spectroscopy has been employed to investigate the influence of microwave fields on the optical properties of Rydberg excitons in Cu₂O. In particular, Gallagher *et al.* (2022) examined optical transitions between the upper valence band and the lowest conduction band (‘yellow’ excitonic series) [8]. Here, we briefly summarise the relevant selection rules and the corresponding spectroscopic signatures.

An excitonic optical transition is allowed when the electric dipole matrix element $\langle \psi_i | e\mathbf{r} | \psi_f \rangle$ is non-zero [18, 51, 57], where ψ_i denotes the crystal ground state and ψ_f the final excitonic state. Since the dipole operator $e\mathbf{r}$ has odd parity [57], the transition is allowed only if the initial and final states have opposite parity.

In Cu₂O, the excitonic wavefunction contains an additional contribution from the envelope function describing the relative motion of the electron-hole pair [41, 51, 57]. Consequently, the parity of the excitonic state depends on both the Bloch functions and the envelope angular momentum L . The envelope function contributes a factor $(-1)^L$, providing an additional degree of freedom in determining optical activity.

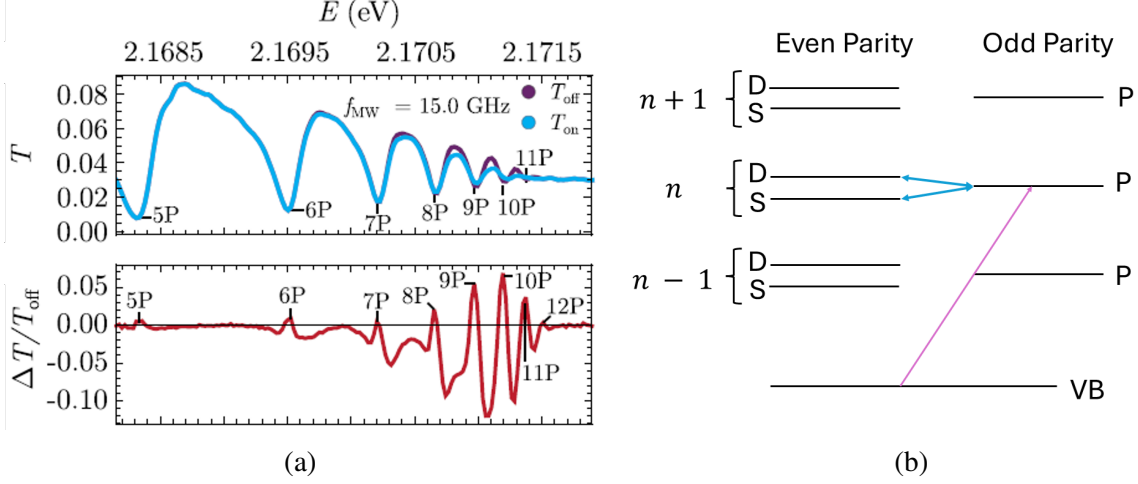


Fig. 4: Coupling of Rydberg excitons in Cu_2O with a microwave field. (a) This figure is adapted from [8]. The microwave field modifies the optical properties of the Rydberg excitons. The absorption spectrum without microwaves is shown in purple, while the spectrum with microwaves is shown in blue. The change in transmitted intensity is shown in the lower panel. It is observed that the transmitted intensity associated with even-parity excitons decreases, whereas that of the odd-parity excitons increases, indicating enhanced absorption in the even-parity excitons. This behaviour can be explained by the mixing of even- and odd-parity excitonic states shown in (b). The mixing causes the even-parity states to acquire characteristics of the odd-parity states.

As discussed in [57], S ($L = 0$) and D ($L = 2$) excitons possess even-parity envelope functions and are therefore dipole-forbidden in one-photon absorption within the electric dipole approximation. In contrast, P ($L = 1$) and F ($L = 3$) excitons have odd parity and are dipole-allowed. In transmission spectra, this results in pronounced minima (strong absorption) at the P and F exciton resonances, while S and D states appear as comparatively weak or peak-like features.

In the experiment described by [8], a mechanically polished thin Cu_2O crystal was mounted between CaF_2 windows in a copper holder and cooled to approximately 4 K using liquid helium. Broadband light was transmitted through the sample and analysed using a grating monochromator. The measurements were repeated under applied microwave fields.

Without microwave excitation, the transmitted intensity exhibits clear minima at the P exciton resonances as shown in Fig. 4(a), consistent with the selection rules. Upon application of microwave fields, the transmission spectrum is modified: the intensity at the S and D resonances decreases, while that at the P resonances increases. For higher principal quantum numbers ($n > 7$), changes exceeding 10% are observed.

In an ideal case, an absorption or transmission spectrum would exhibit a Dirac delta function, indicating that photon absorption occurs only when the photon energy exactly matches the transition energy. However, the transmission spectra shown in Fig. 4(b) exhibit significant broadening. This broadening arises from phonon-assisted transitions to the 1S ortho-exciton of the yellow series [41, 42]. The resulting spectral profile can be described by a Fano lineshape,

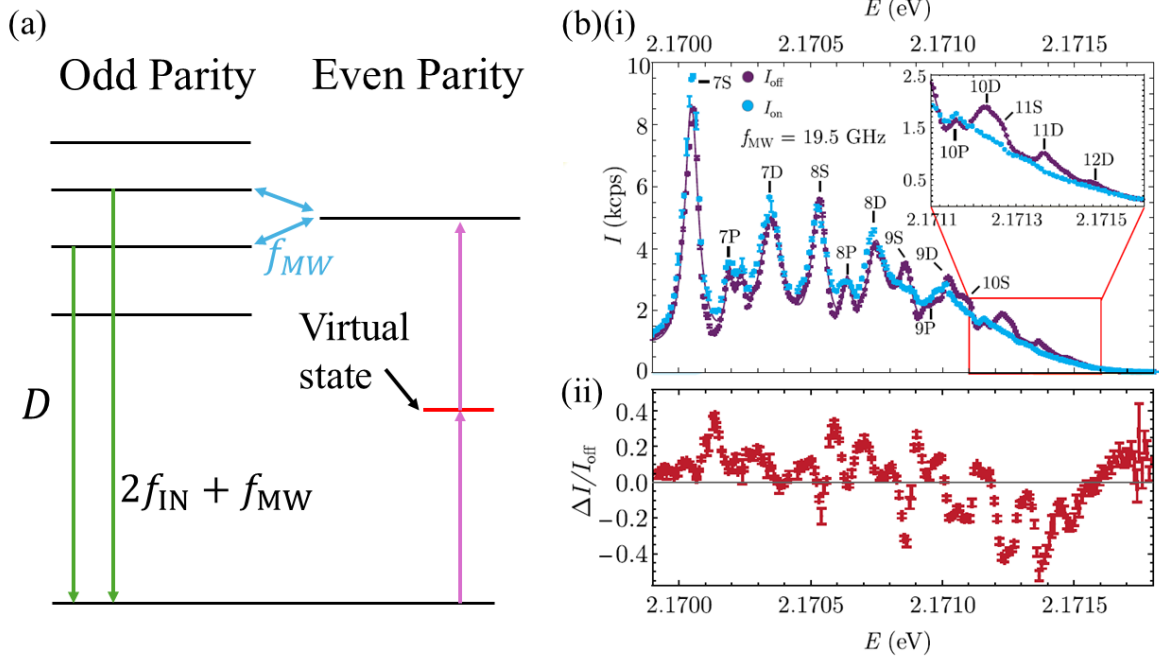


Fig. 5: Second harmonic generation of Cu_2O to access the even-parity excitons. (a) The even-parity excitons are excited through two-dipole process. As the microwave field is applied to the crystal, it couples the even- and off-parity states, allowing the excitons to de-excite and release a photon of frequency $2f_{IN} + f_{MW}$. (b)(i) This figure is adapted from [8]. The initial emitted SHG intensity is shown by the purple dots, while the modified intensity in the presence of the microwave field is indicated by the blue dots. (b)(ii) The relative change in emitted intensity, $\Delta I/I_{off}$, is plotted, where the maximum observed change exceeds 0.4.

which originates from the interference between the spectrally narrow discrete exciton states and the continuum of phonon-assisted transitions [41].

The authors attribute this behaviour to microwave-induced mixing between exciton states of different parity, as illustrated in Fig. 4(b). The microwave field couples nearby Rydberg exciton levels, allowing nominally dipole-forbidden even-parity states to acquire odd-parity character [42], and vice versa. In essence, this measurement shows that the microwave field has a sizeable effect on the optical properties of Rydberg excitons in Cu_2O .

6.2 Second Harmonic Generation

In this section, we discuss another measurement technique used in [8] to probe the coupling between exciton states of opposite parity. As mentioned in Section 6.1, the excitation of S and D excitons is dipole-forbidden in one-photon absorption, and therefore these states do not exhibit strong absorption features. However, even-parity (S and D) Rydberg excitons can be accessed through second harmonic generation (SHG).

Second harmonic generation (SHG) is a $\chi^{(2)}$ nonlinear optical process in which two photons of identical frequency interact within a non-centrosymmetric medium to produce a single photon at twice the frequency [40, 50]. This nonlinear technique offers practical advantages: stray light

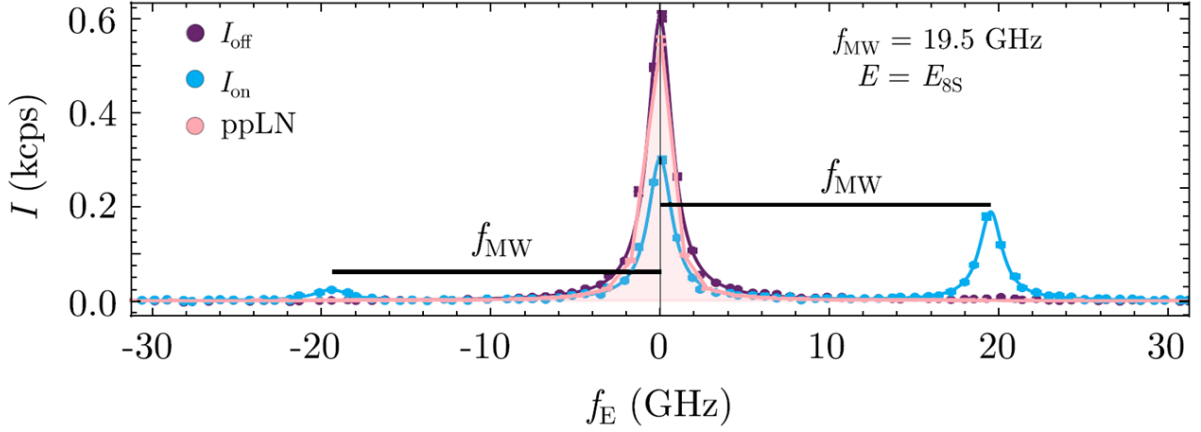


Fig. 6: This figure is adapted from [8]. Resolved emitted second-harmonic generation intensity I using an etalon (a precision optical filter). The intensity of the central peak decreases, and two red and blue sidebands are formed at frequencies f_{MW} away from the central peak.

[66] and strong background absorption from optical phonons [42, 66] can be significantly suppressed.

Cu_2O belongs to the O_h point group [41, 42, 57] and is therefore centrosymmetric. In principle, bulk SHG is forbidden in such materials. However, SHG can occur when the incident light propagates along a low-symmetry crystallographic direction such as [111] [67], or when strain, lattice imperfections, or externally applied fields reduce the symmetry of the crystal.

Although emission from even-parity Rydberg excitons is dipole-forbidden, it can occur via an electric quadrupole process. In the present experiment, even-parity excitons are excited through a two-photon dipole process at frequency f_{IN} , and subsequently emit a photon at twice this frequency, $2f_{\text{IN}}$, via a quadrupole transition. A microwave field with frequency f_{MW} couples nearby exciton states and introduces a new emission pathway at $2f_{\text{IN}} + f_{\text{MW}}$, as illustrated in Fig. 5(a). Consequently, the emitted intensity for exciton states with principal quantum numbers $n = 7$ to $n = 12$ is modified, with a maximum change exceeding 0.4, as shown in Fig. 5(b).

To further investigate the influence of the microwave field, a temperature-tuned etalon (high-precision optical filter) was used to perform high-resolution spectroscopy of the emitted SHG light. By scanning the etalon around $E = E_{8\text{S}}$, a reduction in the central peak intensity was observed as a function of etalon detuning, together with the emergence of two optical sidebands at frequencies $f_E = \pm f_{\text{MW}}$ relative to the central peak, as shown in Fig. 6. The blue sideband corresponds to $+f_{\text{MW}}$, while the red sideband corresponds to $-f_{\text{MW}}$.

The asymmetry in the sideband amplitudes is attributed to differences in excitation energy. The nearest states that mix with the $n\text{S}$ states are higher-energy P states, whereas for the $n\text{D}$ states, the nearest coupled states are lower-energy S states. Consequently, when even-parity excitons are excited, the blue sidebands exhibit greater intensity than the red sidebands.

While the excitation methods described in Sections 6.1 and 6.2 provide viable optical platforms for microwave photon readout, each presents technical challenges. One-photon spec-

troscopy suffers from strong phonon absorption background, whereas SHG is limited by low conversion efficiency due to the centrosymmetric nature of Cu_2O . The former limitation may be mitigated using EIT, while the latter can be addressed by applying external fields to break crystal symmetry [67].

In the following section, we discuss a further technical challenge for Rydberg exciton-based microwave-to-optical conversion, namely the synthetic fabrication of high-quality Cu_2O crystals.

6.3 Limitation of Rydberg Exciton Platform

For Rydberg exciton-based microwave-to-optical conversion (MOC) to be viable for large-scale or industrial applications, reliable and high-quality synthetic fabrication of Cu_2O is essential. At present, however, the fabrication techniques remain insufficiently mature. To date, key experimental advances, including the observation of $n = 30$ yellow excitons [54], spectroscopy of even-parity Rydberg excitons [66], and the microwave coupling experiments described in Section 6 have primarily relied on natural Cu_2O crystals.

Various methods have been developed to fabricate synthetic Cu_2O , including melting [68], electrodeposition [69], chemical vapour deposition [70], and the floating zone method [56]. Here, we briefly review the work reported in [56], in which the optical properties of Rydberg excitons in synthetic Cu_2O were experimentally investigated.

The floating zone method offers several advantages. It avoids thermal strain that may arise during cryogenic cooling when Cu_2O is bonded to a substrate with a different thermal expansion coefficient, and it eliminates the need for a crucible, thereby preventing chemical reactions between molten Cu_2O and container materials. Despite these advantages, the experiment revealed that, for samples of comparable thickness, Rydberg excitons up to $n = 14$ were observable in natural crystals, whereas only up to $n = 10$ were detected in the synthetic samples.

To determine the origin of this reduced maximum principal quantum number, the authors performed photoluminescence and mid-infrared spectroscopy measurements. By comparing the spectra of natural and synthetic samples, they attributed the decrease in observable n to a high density of copper vacancies introduced during mechanical polishing.

The authors suggested several strategies to mitigate this limitation. High-temperature annealing may reduce the concentration of copper vacancies, while alternative surface preparation techniques, such as chemical etching or ion-beam milling, could minimise damage caused by mechanical polishing. Nevertheless, to date, no systematic study has demonstrated significant improvement in the quality of synthetic Cu_2O for high- n Rydberg exciton applications.

7 Outlook

The performance of the MOC platforms discussed in Sections 3.3 and 5 is summarised in Table 2. While these platforms have demonstrated promising results, there remains significant room for improvement. In this section, we discuss several potential developments that could enhance the performance of microwave-optical converters based on Rydberg atoms and Rydberg excitons.

The authors of [10] suggest that the efficiency of ultracold Rydberg atom MOCs could be improved through better matching of the millimetre-wave field (electromagnetic waves with frequencies between the radio-frequency and microwave regimes) to the converter. In particular, concentrating the microwave field into the interaction region using waveguides would enhance the coupling between the microwave field and the Rydberg atoms. For room-temperature Rydberg atom MOCs, the authors of [62] proposed several improvements to reduce non-thermal noise, such as exploring non-collinear configurations of the laser fields and utilising different optical elements within the experimental setup.

In addition to these experimental optimisations, improvements in the coherence properties of Rydberg states have also been proposed. In particular, [48] demonstrated the use of circular Rydberg states with maximal angular momentum. These states exhibit significantly longer lifetimes due to selection rules that suppress spontaneous decay to lower-lying orbitals.

While Rydberg atoms provide an effective platform for microwave-optical conversion and their potential improvements are well understood, their experimental requirements remain incompatible with superconducting qubit platforms. In particular, the laser cooling and trapping techniques required for atomic Rydberg systems cannot easily be integrated with dilution refrigerators used for superconducting circuits. This motivates the exploration of solid-state alternatives such as Rydberg excitons in Cu_2O .

Although a complete Rydberg exciton-based microwave-optical converter has not yet been realised, experimental progress has been encouraging. In particular, the authors in [8] reported a coupling parameter of approximately 0.9 even with a relatively inefficient microwave antenna. This suggests that the strong-coupling regime, where the coupling parameter exceeds unity, may be achievable through improved microwave antenna design and optimisation of the experimental geometry.

	Efficiency (%)	Bandwidth (MHz)
Opto-Electro-Mechanical	47	1.2×10^{-2}
Ferromagnetic Magnon	10^{-8}	2.28
Ultracold Rydberg Atoms	5	4
Room Temperature Rydberg Atoms	3.2	16

Table 2: Summary of the conversion efficiencies and bandwidths of physical platforms for microwave-optical converters

8 Conclusion

This thesis reviews the state-of-the-art microwave-optical converters using different media, including opto-electro-mechanical systems, ferromagnetic magnons, ultracold and room-temperature Rydberg atoms.

In Section 3, we discussed the development of quantum computing and the challenges involved, particularly the decoherence of quantum information when it is transported over large distances. Since optical photons have demonstrated relatively successful long-distance transmission of quantum information, this motivates the development of microwave-optical converters (MOCs) to bridge microwave-based quantum processors and optical quantum networks. We also reviewed several examples of MOCs based on non-Rydberg platforms and discussed the trade-off between conversion efficiency and bandwidth. This limitation motivates the exploration of alternative systems in which a more favourable balance between conversion efficiency and bandwidth can potentially be achieved.

In Section 4, we discussed the properties of Rydberg atoms and Rydberg excitons that can be utilised in microwave-optical converters, such as their large spatial extent and high polarisability. We also examined the use of Rydberg excitons in Cu_2O as the coupling medium between microwave and optical fields. A key advantage of using Rydberg excitons in Cu_2O is their relative stability, which arises from the comparatively large exciton binding energy.

In Section 5, we discussed the MOC via ultracold Rydberg atoms through six-wave mixing. The experimental setup involved a closed loop of the four auxiliary optical fields and two microwave fields and the conversion occurred through a nonlinear process where the frequencies of the optical and microwave fields combine to form a new frequency. We also discussed the MOC via room temperature Rydberg atoms using a similar setup with only one microwave field. We then compare between the choice of ultracold and room temperature Rydberg atoms.

In Section 6, we discussed the methodology used in investigating the coupling between Rydberg excitons in Cu_2O and microwave field, particularly one-photon spectroscopy and second harmonic generation. The authors reported a large coupling parameter despite having a relatively inefficient microwave antenna. Therefore, strong-coupling regime is achievable with an improved microwave antenna design.

The development of microwave-optical converters is essential for enabling quantum communication between microwave-based quantum processors and optical quantum networks. Although these technologies are still in their early stages of development, significant progress has been made across multiple platforms. In particular, Rydberg-based systems offer promising opportunities due to their strong coupling to microwave fields and favourable nonlinear optical properties. Continued progress in these directions may ultimately enable microwave-optical converters to serve as a key interface linking superconducting quantum processors with large-scale optical quantum communication networks.

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9 Lay Summary for General Audience

Since the twentieth century, physicists have been developing quantum computers to solve problems that require excessive computational time for classical computers. While classical computers use bits as the fundamental operating units, which can only take the value 0 or 1, quantum computers use qubits, which operate at microwave frequencies and can exist in a combination of both states at the same time, a property known as quantum superposition.

To allow quantum computers to form networks, they must be able to communicate with each other. In many quantum processors, information is carried by microwave photons. However, microwave photons are easily lost as heat during transmission, making long-distance communication difficult. Optical photons, on the other hand, can travel long distances with relatively little loss. This motivates the development of microwave–optical converters (MOCs), which convert signals between microwave and optical frequencies.

This thesis reviews the development of MOCs, focusing particularly on systems based on Rydberg atoms, which are atoms excited to very high energy levels. We also discuss the potential of Rydberg excitons in semiconductors as an alternative platform. Current experiments show promising results, suggesting that Rydberg-based systems may provide an effective interface between microwave quantum processors and optical communication networks.

(199 words)

10 Statement of Generative AI

The only generative AI tool used for this report was ChatGPT. It was used solely for grammar and phrasing adjustments, without generating new technical content.